MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE EHU-31A/A

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for single-pointer, self-balancing, potentiometer-type temperature indicator type EHU-31A/A ac white lighted (19 pin receptacle).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-I-27247	Indicator, Temperature, Synchro.
MIL-C-81703	Connector, Electric, Circular, Miniature, Rack and Panel or Push-Pull Coupling, Environmental Resisting.
MIL-I-83262	Indicator, Temperature, Thermocouple, Hermetically Sealed, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-595	Color.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-810	Environmental test Methods and Engineering Guidelines.
MS3106	Connector, Plug, Electric Straight.
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument.
MS33639	Case, Instrument, Clamp-Mounted, Aircraft.
MS33678	Connector, Receptacle, Electric, Integral Mounting.

* (Copies of specifications and standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.1.2 and 6.2).

3.2 Power consumption. The indicator steady-state power consumption shall not exceed 7Va exclusive of the lighting circuit.

3.2.1 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of at least 2,000 hours at a 90 percent confidence level.

3.3 Design and construction. The indicator shall incorporate a self-balancing potentiometer-type circuit for temperature measurement. Other major components of the indicator are:

a. An input isolation-type power transformer

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c. 115V 400 Hz syncho transmitter, clifton precision part number CGH10-NS-9 or equal. The indicator shall supply a synchro repeater output which shall drive an EHU-11A/A temperature indicator in accordance with MIL-I-27247. the motor shall permit movement of the pointer by direct coupling and without ther use of gears or pulleys, unless a geared design is specifically approved.

3.3.1 Power supply. The power supply shall be in accordance with MIL-I83262 for ac powered indicators.

a. Scale error not exceeding the tolerance as specified in table I.

b. Friction error not exceeding 2°C over a temperature range of 510° to 790°C.

c. Position error not exceeding 1°C over a temperature range of 510° to 790°C.

d. Case leakage not exceeding 0°1 micron cubic foot per hour.

e. Warmup and pointer travel from 200° to 1,000°C within 3 seconds at room temperature and 6 seconds at -40° and +†00°C.

f. Instaneous indication change from 500° to 850°C in not more than 2 seconds at room temperature and 2.5 seconds at -55° to +100°C with any combination of voltages and frequencies from 95 to 135V and from 320 to 480 Hz.

g. When the signal is increased or decreased 0.08 mv a change of 2° +1°C indication shall be noted at points between 510° and 790°C.

h. Withstand temperatures ranging from -65° to +100°C.

i. Withstand an alternate input of -3 and +52 mv for a period of 30 minutes at each condition for eight cycles with the ambient temperature of 50°C.

j. Withstand an alternate input of -3 and +52 mv for a period of 30 minutes at each condition for eight cycles with the ambient temperature of 50°C.

k. Indication change not exceeding 2°C when a lead resistance change from 5 to 200 ohms is made at specified test points between 600° and 790°C.

l. Indication steady-state change not exceeding 2°C when either the chromel or alumel conductor is grounded separately at any point to the ground side (low potential side) of the 115V 400 Hz power supply when the indicator is operated in the range of 600° to 790°C.

m. Withstand the effects of beat frequencies and stray ac signals in accordance with MIL-I-83262.

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n. Withstand pressures ranging from 30 inches Hg down to 0.309 inch Hg, (approximately sea level to 100,000 feet).

o. Withstand frequency variations ranging from 320 to 480 Hz, voltage variations from 95 to 135V steady-state and voltage surges to 195 ±5V.

p. Withstand eight cycles of alternate immersions in tap water at 85° ±2°C and 5° ±2°C.

q. Withstand a mechanical shock of 10g for 11 ±1 milliseconds with no damage or malfunction of any component of the indicator.

r. Exposure to salt sea atmosphere for a period of 48 hours.

TABLE I. Test tolerances, EHU-31A/A.

Test Points °C	Scale Error at Room Temperature ±°C	Scale Error at Low Temperature ±°C	Scale Error at High Temperature ±°C	Scale Error During Reliability Test ±°C	Friction Error °C
100	15	20	20	20	---
200	15	20	20	20	---
400	15	20	20	20	---
510	5	10	10	10	2
600	5	10	10	10	2
700	5	10	10	10	2
790	5	10	10	10	2
900	15	20	20	20	---
1,000	15	20	20	20	---
1,100	15	20	20	20	---
1,200	15	20	20	20	---

3.5 Case. The indicator case shall be hermetically sealed in accordance with MS33639 for the 2-inch diameter size. The case length shall be 7 ±0.25 inches exclusive of the electrical receptacle. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, color number 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental test conditions specified herein. The hermetic seal shall not be accomplished by use of materials that are affected by the action of any atmosphere to which the case may be subjected. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than .03125 inch. The case shall be so designed that the internal mechanisms may be removed from the case, replaced, and the case resealed without the use of special tools unless they are approved by the procuring activity.

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3.6 Dial. The indicator dial shall be as shown on figure 1 and as otherwise specified herein. A minimum diameter of 1.75 inches shall be maintained across the outer ends of the scale graduations. The total scale arc shall be 340 ±2° within the contracted and expanded portions of the scales the scale shall be within 0.5 angular degree of precise graduations. The background of the dial shall be durable lusterless black consistent with integral lighting requirements. An electrical-zero reference mark shall be located at the outer end of the 650°C scale graduation as shown on figure 1.

3.6.1 Power-on display. The EHU-31A/A indicator shall indicate application of input power by movement of the pointer to uncover the word ON (see figure 1). The word ON shall be in dayglo signal green letters, or equal, and shall be lighted to a brightness level of 1 ±0.5 foot-lamberts. When power is removed the pointer shall return to a position over and shall completely cover the word ON within 3 seconds regardless of original pointer position. When power is applied, the pointer shall move to the proper temperature indication. If the temperature is below 200°C, the pointer shall move enough to uncover the word ON.

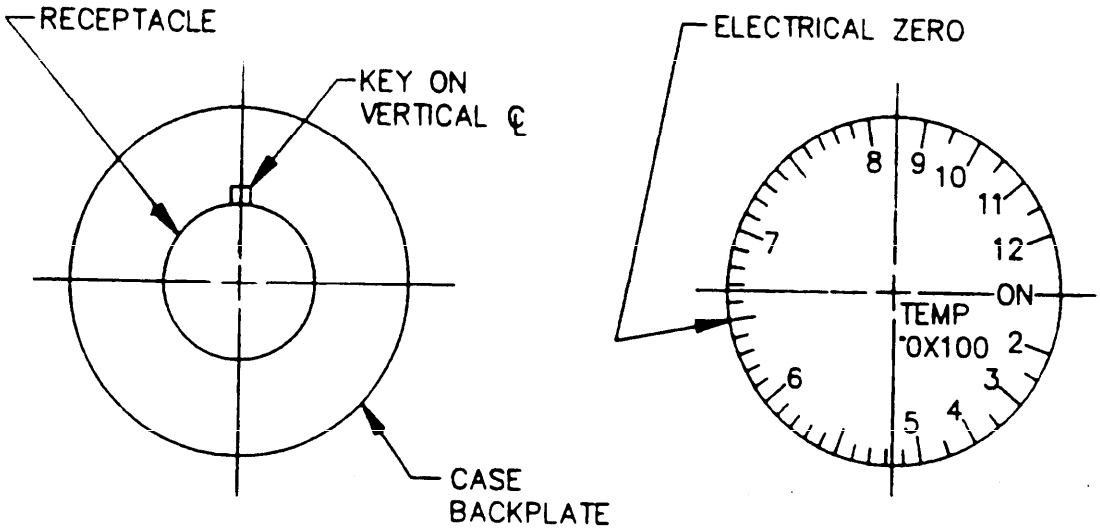
3.7 Pointer. The pointer shall be in accordance with MS33585-8, except the tip of the pointer shall extend to the inner ends of the major scale graduations ±0.016 inch. The length of the lighted portion of the pointer shall be not less than 0.60 inch.

3.8 Marking dimensions and colors. The dial and pointer colors shall be in accordance with FED-STD-595. The dimensions of the numerals and letters shall be as follows:

	Height or Length Inch + 0.005	Width of Line or Graduation Inch + 0.005	Colors
Numerals	0.156	0.031	37875
50°C graduations (500° to 800°C)	0.156	0.031	37875
100°C graduations	0.156	0.031	37875
50°C graduations (except 500° to 800°C)	0.125	0.020	37875
Lettering temperature °C x 100	0.094	0.020	37875
10°C graduations	0.125	0.020	37875
Background of dial	---	---	37038
Pointer	---	---	37875
Electrical zero Reference mark	0.031	0.010	37875
Word ON	0.094	0.015	(see 3.6.1)

3.9 Electrical receptacle. The EHU-31A/A indicator shall be provided with a hermetically sealed electrical receptacle in accordance with MIL-C-81703 type MS3463H19A50P or deutsch type DM5600-19P, part number 14375-0025, or equal. Pin number 13 shall be made of chromel material and pin number 15 shall be made of alumel material.

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EHU-31A/A DIAL

NOTES:

TOTAL LENGTH OF E-Z MARK PLUS 650°C GRAD TO BE 0.156 ±0.0005. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ANGLES ±0° 15'.

FIGURE 1. Dial and case backplate.

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3.9.1 Wiring. The receptacle shall be wired internally for operation in accordance with external connections which are based upon clockwise rotation of a cynchro rotor shaft for increasing temperatures. If gears are used and pointer rotation is opposite to that of synchro shaft rotation, synchro transmitter connections to pins 2 and 3 of the receptacle shall be reversed to cancel the effects of the gear reversal.

3.9.1.1 EHU-31A/A. The following connections shall be used for the EHU-31/A:

Pin	External Connection
1	115V 400 Hz external ground internally connected to pin 4
2	Synchro receiver stator S3
3	Synchro receiver stator S1
4	Ground for synchro receiver S2 and R2
5	115V 400 Hz to receive rotor R1
6	Open
7	Open
8	Open
9	115V 400 Hz input
10	5V lighting input
11	Open
12	Open
13	Chromel lead wire
15	Alumel lead wire
14, 16, 17, 18, 19	Open

3.9.2 Grounding of wire. The case shall be grounded by connecting a wire from the terminal of pin 1 of the receptacle to a lug on the inner surface of the backplate of the indicator.

3.10 System connections. The EHU-31A/A indicators shall be connected as shown on figure 2 for system operation.

3.11 Excluded components. The design and fabrication of the indicator shall not include the following components.

a. Electromechanical choppers.

b. Gears for motor movement transfer purposes unless specifically approved.

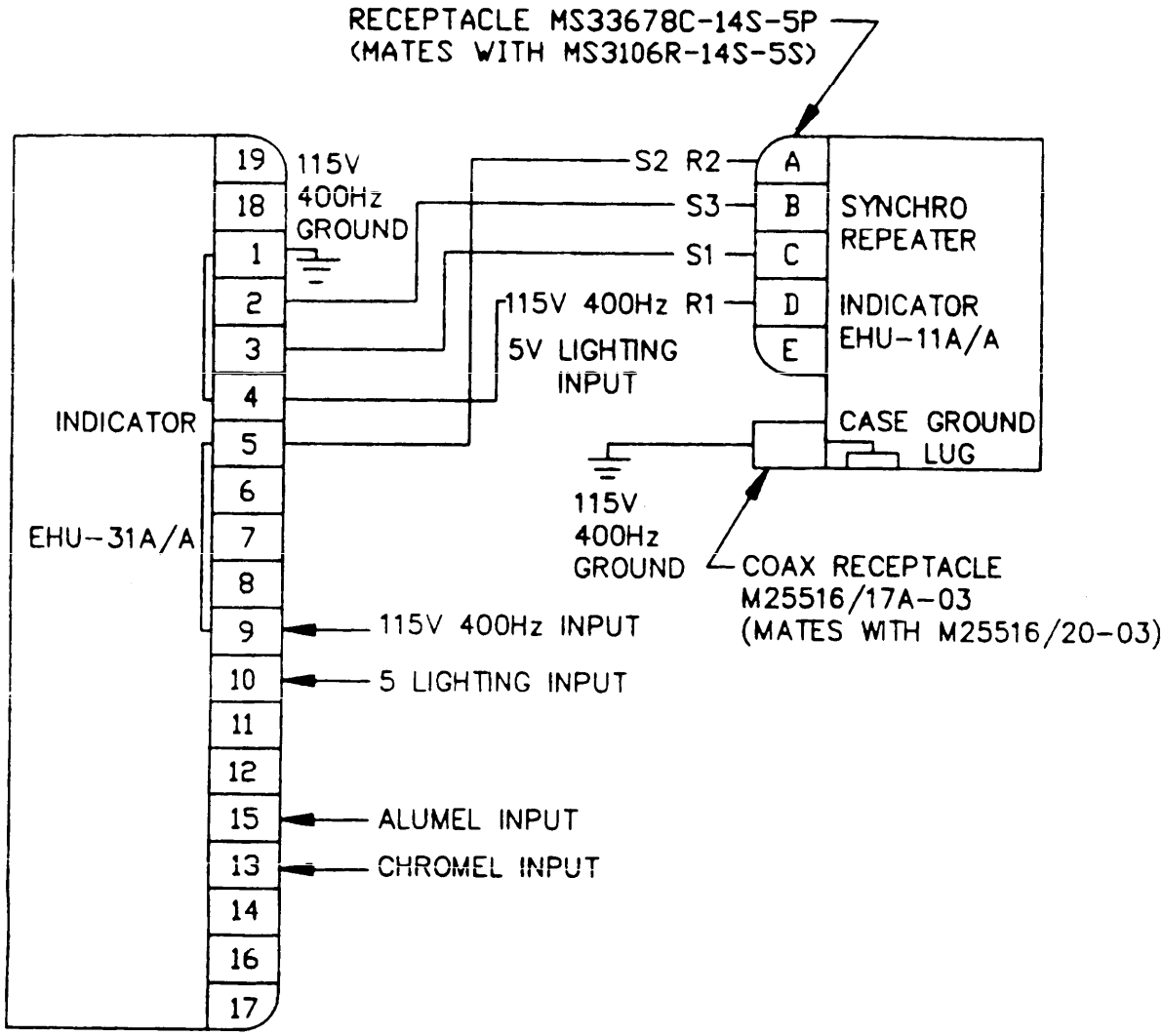
c. Plastic packaged integrated circuits.

3.12 Weight. The weight of the indicator shall not exceed 1.50 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, inspection and test.

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RECEPTACLE, DEUTCH TYPE DM5600-19P OR EQUAL
(MATES WITH DEUTSCH TYPE DM9700-19S)

NOTE: PINS AND SOCKETS NO 13 AND NO 15 SHALL BE MADE OF CHROMEL AND ALUMEL, RESPECTIVELY

FIGURE 2. EHU-31A/A Indicator wiring.

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. the inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 First article provisions.

4.1.2.1 First article inspection. The first article inspection shall be in accordance with MIL-I-83262/1 and as specified herein. Unless otherwise specified herein, the contractor shall perform the inspection and submit certification that the indicators conform to the requirements of this specification. A copy of the first article test results shall be furnished to the Government.

4.1.2.2 Sampling, inspection and test.

4.1.2.3 Sampling. Sampling shall be in accordance with MIL-I-83262.

4.1.3 Test methods. Unless otherwise specified herein, all test methods and procedures shall be in accordance with MIL-I-83262.

4.1.3.1 Power supply. Unless otherwise specified, the indicator shall be tested with 115V $\pm$2V single-phase 400 $\pm$5 Hz ac power with 4.5 $\pm$0.5V ac or dc applied to the lighting circuit.

4.1.3.2 Electrical-zero. The indicator shall be tested in accordance with MIL-I-83262.

4.1.3.3 Scale error and friction. The friction error test shall be in accordance with MIL-I-83262 and shall be inducted at 510°, 600°, 700°, and 790°C.

4.1.3.4 Position error. This test shall be conducted at the 600°, 700°, and 790°C test points.

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4.1.3.5 Response time. Millivoltage signals and power shall be applied to provide a stable indication at the low end of the expanded scale. The millivoltage shall then be increased to produce an instantaneous change equivalent to 350°C. The time required for the indicator to fully respond to the 350°C change shall be noted and shall not exceed 2 seconds. This test shall not exceed 2 seconds. This test shall be conducted using the following combinations of input voltage and frequencies: 115V 400 Hz, 95V 320 Hz, 95V 480 Hz, 135V 320 Hz and 135V 480 Hz. The test shall be repeated going from the high end of the expanded scale to the low end of the expanded scale. The same tolerance shall apply.

4.1.3.6 Sensitivity. The procedure shall be repeated at the 600° and 790°C test points.

4.1.3.7 Low temperature compensation. The test will be in aaccordance with MIL-I-83262 except that before power is reapplied the temperature of the chamber shall be raised to -40°C before conducting the specified tests.

4.1.3.8 Low temperature exposure. The test wil be in accordance with MIL-I-83262 except that the temperature shall be raised to -40°C and maintained for a period of 24 hours.

4.1.3.9 Thermocouple lead resistance effect. This test shall be conducted at the 600°, 700°, and 790°C test points.

4.1.3.10 Grounding of thermocouple extension wires. Scale error shall be determined at the 600° and 790°C test points and the response time shall be determined in the 500° to 850°C range on increasing temperature.

4.1.3.11 Vibration. The indicator shall be subjected to vibration in accordance with MIL-STD-810, method 514.3, equipment category 4, figure 514.3-25. Pointer oscillation shall not exceed 5°C double amplitude. The test point shall be 700°C with power applied. Following the vibration test, the indicator shall pass the scale error and friction test of MIL-I-83262. No screws or other parts shall become loosened or damaged as a result of this test.

5. PACKAGING

5.1 Preservation-packaging. Preservation -packaging shall be in accordance with MIL-I-83262.

5.2 Marking. In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.1 Nomenclature. The nomenclature shall be as follows:

INDICATOR, TEMPERATURE,
THERMOCOUPLE, SELF-BALANCING
POTENTIOMETER EHU-31A/A

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6. NOTES

6.1 All noted specified in MIL-I-83262 are applicable to this specification.

6.2 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The first article should consist of three units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.3 Subject term (key word) list.

Indicator

Temperature

Thermocouple

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

(Project 6685-F798)

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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